

3.7 How Science Works

How Science Works is an underpinning set of concepts and is the means whereby students come to understand how scientists investigate scientific phenomena in their attempts to explain the world about us. Moreover, How Science Works recognises the contribution scientists have made to their own disciplines and to the wider world.

Further, it recognises that scientists may be influenced by their own beliefs and that these can affect the way in which they approach their work. Also, it acknowledges that scientists can and must contribute to debates about the uses to which their work is put and how their work influences decision-making in society.

In general terms, it can be used to promote students' skills in solving scientific problems by developing an understanding of:

- the concepts, principles and theories that form the subject content
- the procedures associated with the valid testing of ideas and, in particular, the collection, interpretation and validation of evidence
- the role of the scientific community in validating evidence and also in resolving conflicting evidence.

As students become proficient in these aspects of How Science Works, they can also engage with the place and contribution of science in the wider world. In particular, students will begin to recognise:

- the contribution that scientists, as scientists, can make to decision-making and the formulation of policy
- the need for regulation of scientific enquiry and how this can be achieved
- how scientists can contribute legitimately in debates about those claims which are made in the name of science.

An understanding of How Science Works is a requirement for this specification and is set out in the following points which are taken directly from the GCE AS and A Level subject criteria for science subjects. Each point is expanded in the context of Physics. The specification references given illustrate where the example is relevant and could be incorporated.

A	Use theories, models and ideas to develop and modify scientific explanations Scientists use theories and models to attempt to explain observations. These theories or models can form the basis for scientific experimental work. Scientific progress is made when validated evidence is found that supports a new theory or model. Candidates should use historical examples of the way scientific theories and models have developed and how this changes our knowledge and understanding of the physical world. <i>Examples in this specification include:</i> • <i>Galileo deduced from his inclined plane experiment that falling objects accelerate. Newton later explained why and showed that freely-falling objects have the same acceleration. (AS Unit 2 §3.2.1)</i> • <i>The kinetic theory of gases explains the experimental gas laws. (A2 Unit 5 §3.5.3)</i>
B	Use knowledge and understanding to pose scientific questions, define scientific problems, present scientific arguments and scientific ideas Scientists use their knowledge and understanding when observing objects and events, in defining a scientific problem and when questioning their own explanations or those of other scientists. Scientific progress is made when scientists contribute to the development of new ideas, materials and theories. Candidates will learn that: • a hypothesis is an untested idea or theory based on observations • predictions from a hypothesis or a theory need to be tested by experiment • if a reliable experiment does not support a hypothesis or theory, the hypothesis or theory must be changed. <i>Examples in this specification include:</i> • <i>Many opportunities permeating throughout the Investigative and Practical Skills units (Unit 3 & 6)</i>
C	Use appropriate methodology, including ICT, to answer scientific questions and solve scientific problems Observations ultimately lead to explanations in the form of hypotheses. In turn, these hypotheses lead to predictions that can be tested experimentally. Observations are one of the key links between the 'real world' and the abstract ideas of science. Once an experimental method has been validated, it becomes a protocol that is used by other scientists. ICT can be used to speed up, collect, record and analyse experimental data. Candidates will know how to: • plan or follow a given plan to carry out an investigation on topics relevant to the specification • identify the dependent and independent variables in an investigation and the control variables • select appropriate apparatus and methods, including ICT, to carry out reliable experiments relevant to topics in the specification • choose measuring instruments according to their sensitivity and precision. <i>Examples in this specification include:</i> • <i>Many opportunities permeating throughout the Investigative and Practical Skills units (Unit 3 & 6)</i>

	Carry out experimental and investigative activities, including appropriate risk management, in a range of contexts Scientists perform a range of experimental skills that include manual and data skills (tabulation, graphical skills etc). Scientists should select and use equipment that is appropriate when making accurate measurements and should record these measurements methodically. Scientists carry out experimental work in such a way as to minimise the risk to themselves, to others and to the materials, including organisms, used. D Candidates will be able to: • follow appropriate experimental procedures in a sensible order • use appropriate apparatus and methods to make accurate and reliable measurements • identify and minimise significant sources of experimental error • identify and take account of risks in carrying out practical work. <i>Examples in this specification include:</i> • <i>Many opportunities permeating throughout the Investigative and Practical Skills units (Unit 3 & 6)</i>
	Analyse and interpret data to provide evidence, recognising correlations and causal relationships Scientists look for patterns and trends in data as a first step in providing explanations of phenomena. The degree of uncertainty in any data will affect whether alternative explanations can be given for the data. Anomalous data are those measurements that fall outside the normal, or expected, range of measured values. Decisions on how to treat anomalous data should be made only after examination of the event. In searching for causal links between factors, scientists propose predictive theoretical models that can be tested experimentally. When experimental data confirm predictions from these theoretical models, scientists become confident that a causal relationship exists. E Candidates will know how to: • tabulate and process measurement data • use equations and carry out appropriate calculations • plot and use appropriate graphs to establish or verify relationships between variables • relate the gradient and the intercepts of straight line graphs to appropriate linear equations. <i>Examples in this specification include:</i> • <i>Many opportunities permeating throughout the Investigative and Practical Skills units (Unit 3 & 6)</i>

F	Evaluate methodology, evidence and data, and resolve conflicting evidence The validity of new evidence, and the robustness of conclusions that stem from them, is constantly questioned by scientists. Experimental methods must be designed adequately to test predictions. Solutions to scientific problems are often developed when different research teams produce conflicting evidence. Such evidence is a stimulus for further scientific investigation, which involves refinements of experimental technique or development of new hypotheses. Candidates will be able to: • distinguish between systematic and random errors • make reasonable estimates of the errors in all measurements • use data, graphs and other evidence from experiments to draw conclusions • use the most significant error estimates to assess the reliability of conclusions drawn. <i>Examples in this specification include:</i> • <i>Many opportunities permeating throughout the Investigative and Practical Skills units (Unit 3 & 6)</i>
G	Appreciate the tentative nature of scientific knowledge Scientific explanations are those that are based on experimental evidence which is supported by the scientific community. Scientific knowledge changes when new evidence provides a better explanation of scientific observations. Candidates will be able to understand that scientific knowledge is founded on experimental evidence and that such evidence must be shown to be reliable and reproducible. If such evidence does not support a theory the theory must be modified or replaced with a different theory. Just as previous scientific theories have been proved inadequate or incorrect, our present theories may also be flawed. <i>Examples in this specification include:</i> • <i>Antiparticles were predicted before they were discovered. (AS Unit 1 §3.1.1)</i> • <i>Rutherford's alpha scattering experiment led to the nuclear model of the atom even though it was carried out to test Thompson's model of the atom. (A2 Unit 5 §3.5.1)</i>
H	Communicate information and ideas in appropriate ways using appropriate terminology By sharing the findings of their research, scientists provide the scientific community with opportunities to replicate and further test their work, thus either confirming new explanations or refuting them. Scientific terminology avoids confusion amongst the scientific community, enabling better understanding and testing of scientific explanations. Candidates will be able to provide explanations using correct scientific terms, and support arguments with equations, diagrams and clear sketch graphs when appropriate. The need for answers to be expressed in such a way pervades the written papers and the ISA. Furthermore, questions requiring extended writing will be set in which marks may be reserved for demonstrating this skill. <i>Examples in this specification include:</i> • <i>Many opportunities through the assessment of questions requiring extended prose which are evident throughout each of the assessment units in the specification.</i>

I	Consider applications and implications of science and appreciate their associated benefits and risks Scientific advances have greatly improved the quality of life for the majority of people. Developments in technology, medicine and materials continue to further these improvements at an increasing rate. Scientists can predict and report on some of the beneficial applications of their experimental findings. Scientists evaluate, and report on, the risks associated with the techniques they develop and applications of their findings. Candidates will be able to study how science has been applied to develop technologies that improve our lives but will also appreciate that the technologies themselves pose significant risks that have to be balanced against the benefits. <i>Examples in this specification include:</i> • <i>Superconductors are used to make very powerful magnets which are used in MRI scanners. (AS Unit 1 §3.1.3)</i> • <i>A nuclear reactor is a reliable source of electricity and does not emit greenhouse gases but its radioactive waste must be processed and stored securely for many years. (A2 Unit 5 §3.5.2)</i>
J	Consider ethical issues in the treatment of humans, other organisms and the environment Scientific research is funded by society, either through public funding or through private companies that obtain their income from commercial activities. Scientists have a duty to consider ethical issues associated with their findings. Individual scientists have ethical codes that are often based on humanistic, moral and religious beliefs. Scientists are self-regulating and contribute to decision making about what investigations and methodologies should be permitted. Candidates will be able to appreciate how science and society interact. They should examine how science has provided solutions to problems but that the solutions require society to form judgements as to whether the solution is acceptable in view of moral issues that result. Issues such as the effects on the planet, and the economic and physical well-being of the living things on it should be considered. <i>Examples in this specification include:</i> • <i>Secure transmission of data is important if people are to be confident that personal data cannot be intercepted in transmission. (AS Unit 2 §3.2.3)</i> • <i>In the Second World War, scientists on both sides were in a race to build the first atom bomb. (A2 Unit 5 §3.5.2)</i>

K	Appreciate the role of the scientific community in validating new knowledge and ensuring integrity The findings of scientists are subject to peer review before being accepted for publication in a reputable scientific journal. The interests of the organisations that fund scientific research can influence the direction of research. In some cases the validity of those claims may also be influenced. Candidates will understand that scientists need a common set of values and responsibilities. They should know that scientists undertake a peer-review of the work of others. They should know that scientists work with a common aim to progress scientific knowledge and understanding in a valid way and that accurate reporting of findings takes precedence over recognition of success of an individual. Similarly, the value of findings should be based on their intrinsic value and the credibility of the research. <i>Examples in this specification include:</i> • <i>The supposed discovery of cold fusion was rejected after other scientists were unable to reproduce the discovery. (A2 Unit 5 §3.5.2)</i> • <i>The experimental discovery of electron diffraction confirmed the dual nature of matter particles, first put forward by de Broglie as a hypothesis several years earlier. (AS Unit 1 §3.1.2)</i>
	Appreciate the ways in which society uses science to inform decision making Scientific findings and technologies enable advances to be made that have potential benefit for humans. In practice, the scientific evidence available to decision makers may be incomplete. Decision makers are influenced in many ways, including by their prior beliefs, their vested interests, special interest groups, public opinion and the media, as well as by expert scientific evidence. Candidates will be able to appreciate that scientific evidence should be considered as a whole. They should realise that new scientific developments inform new technology. They should realise the media and pressure groups often select parts of scientific evidence that support a particular viewpoint and that this can influence public opinion which in turn may influence decision makers. Consequently, decision makers may make socially and politically acceptable decisions based on incomplete evidence. <i>Examples in this specification include:</i> • <i>Electric cars may replace petrol vehicles if batteries giving a greater range than at present are developed. Until then, car buyers are unlikely to be persuaded to buy electric cars. (AS Unit 1 §3.1.3)</i> • <i>Satellite tracking for purposes such as road pricing may be implemented without adequate trials because of pressure group influence. (A2 Unit 4 §3.4.2)</i>